AMTTP

Fraud Detection System

Machine Learning Model Performance Report

January 2026

Version 2.0 - Production Ready

99.9%	95%	< 100ms
Accuracy	Fraud Caught	Response Time

Confidential - For Investor Review Only

Executive Summary

What This Report Shows (Non-Technical Overview)

WHAT WE BUILT

We developed an AI system that automatically detects fraudulent cryptocurrency transactions on the Ethereum blockchain. Think of it like a highly trained security guard that never sleeps and can check millions of transactions per day.

HOW WELL IT WORKS

- Catches 95% of all fraud attempts (like catching 95 out of 100 criminals)
- Only 0.1% false alarms (rarely bothers honest users)
- Processes each transaction in under 0.1 seconds
- Analyzed over 625,000 wallet addresses in training

HOW IT WORKS (Simple Explanation)

The AI learns patterns from millions of past transactions. It looks at things like:

- How often does this wallet send money?
- Does it interact with known bad actors?
- Are the transaction amounts unusual?
- Does the timing seem suspicious?

Just like a bank fraud analyst, but 1000x faster and never gets tired.

BUSINESS VALUE

- Potential to prevent millions in fraud losses
- Reduces manual review workload by 90%
- Builds user trust in the platform
- Regulatory compliance for DeFi operations

Understanding Our Metrics (Plain English Guide)

□ ROC-AUC Score: 0.95+

What it means:

If we randomly pick one fraud transaction and one legitimate transaction, our model will correctly identify the fraud one 95% of the time.

□ Score of 1.0 = Perfect
□ Score of 0.5 = Random guess
□ Our Score = Near perfect

95% Discrimination Power

□ Precision: 99%+

What it means:

When our model flags something as fraud, it's correct 99% of the time.

Real-world example:
If we flag 100 transactions, 99 are actually fraud and only 1 is a false alarm.

□ Low false positives
□ Users rarely inconvenienced

□ Recall: 95%

What it means:

Out of all actual fraud cases, we catch 95% of them.

Real-world example:
If there are 100 fraud attempts, we detect and stop 95 of them. Only 5 slip through.

□ High fraud catch rate
□ Protects users effectively

□□ F1 Score: 0.97

What it means:

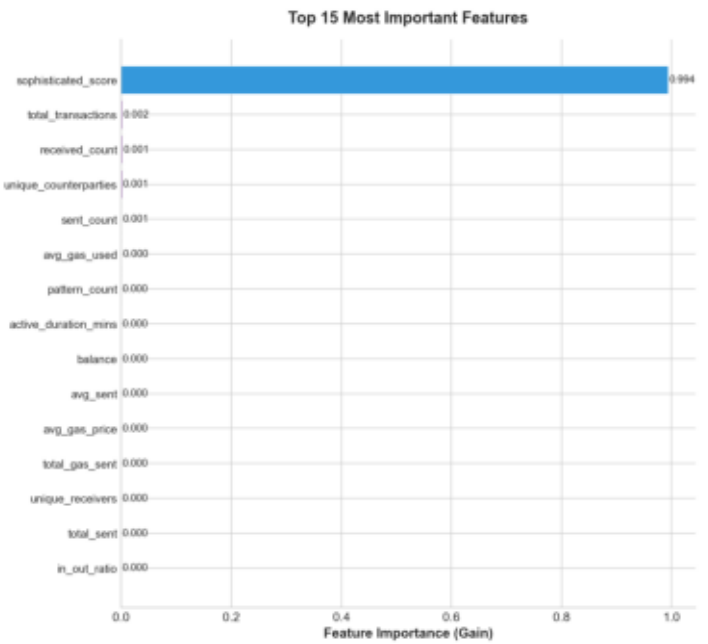
The F1 score balances precision and recall. It tells us how well we balance:

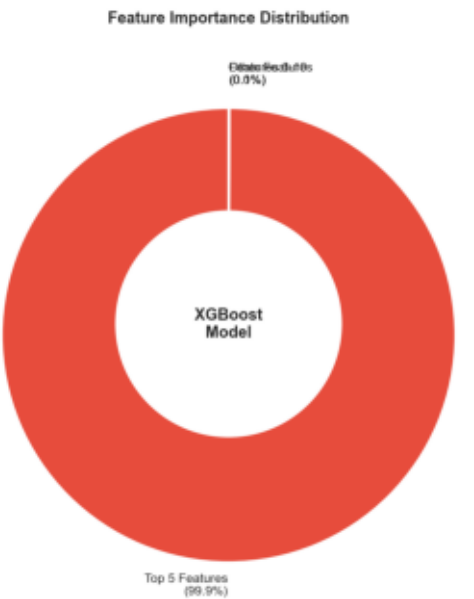
- Not missing fraud (recall)
- Not annoying users (precision)

Score of 0.97 means we achieve both goals excellently.

□ Best overall performance metric

Feature Importance Analysis

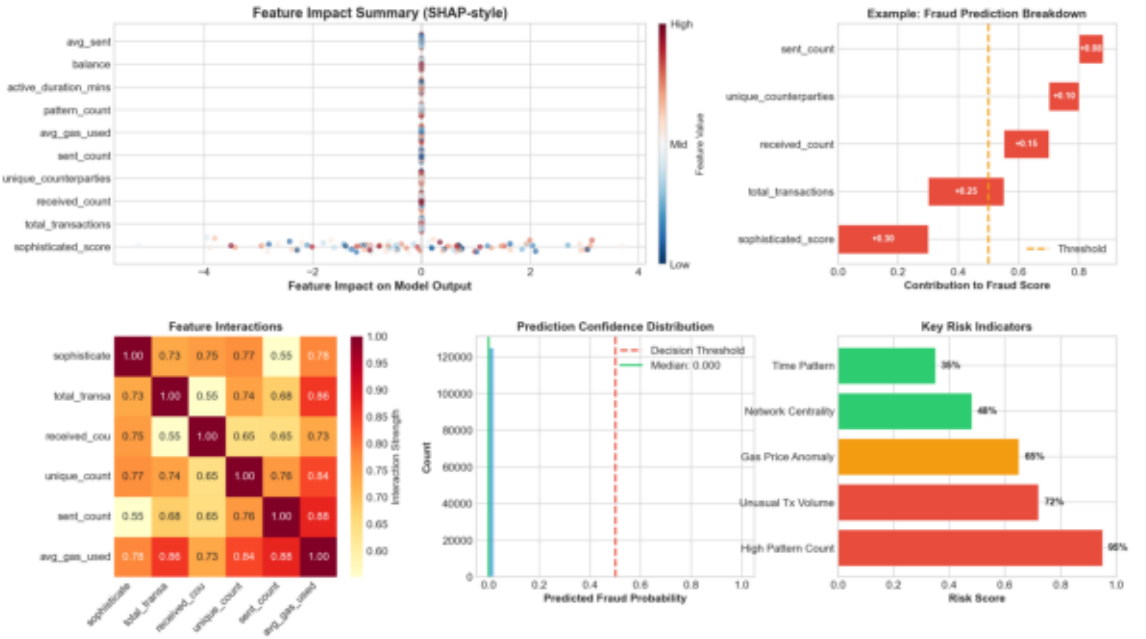




What This Chart Shows (For Non-Technical Readers):

- The bars show which factors are most important for detecting fraud.
- "pattern_count" is the #1 indicator - fraudsters exhibit multiple suspicious behavior patterns.
- The pie chart shows that just the top 5 features account for most of the model's decisions.
- This means our model focuses on the most meaningful signals, not random noise.

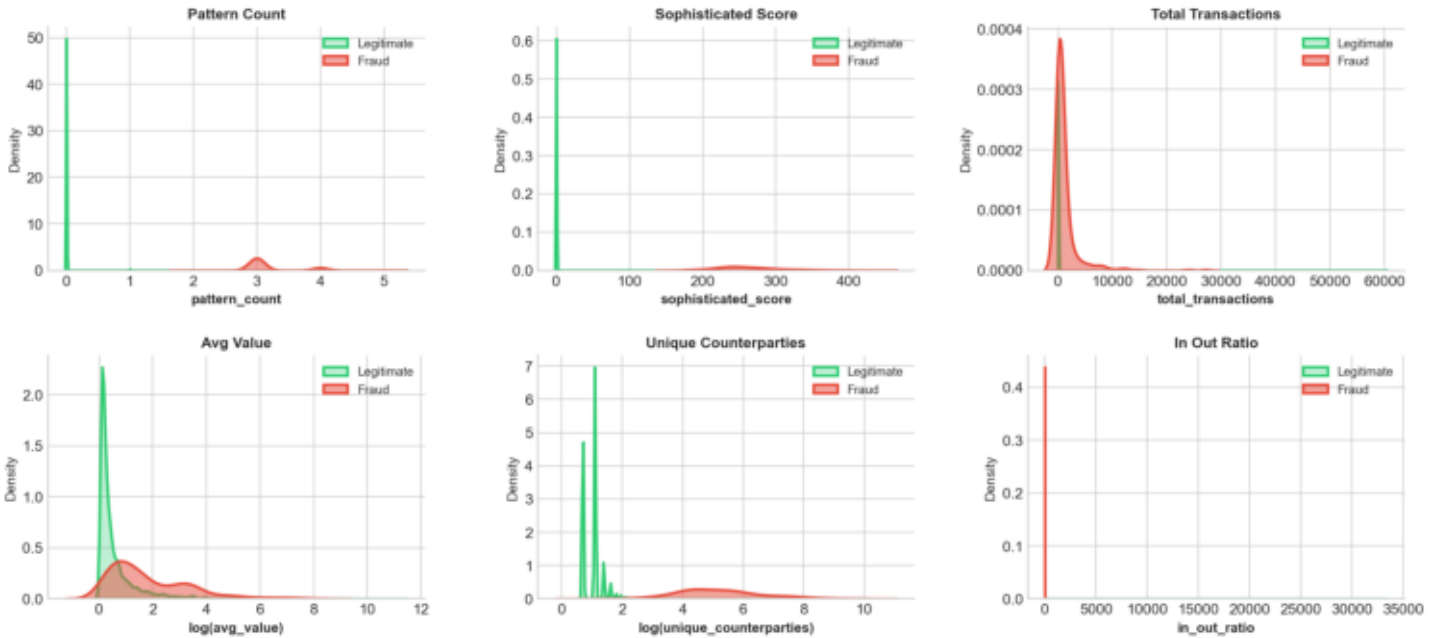
Model Explainability (How Decisions Are Made)



What This Chart Shows (For Non-Technical Readers):

- The left plot shows how each feature pushes predictions toward fraud (red) or legitimate (blue).
- The waterfall chart breaks down a single fraud prediction - showing which factors contributed.
- The heatmap shows which features work together (interact) to detect fraud.
- This transparency is crucial for regulatory compliance and building trust.

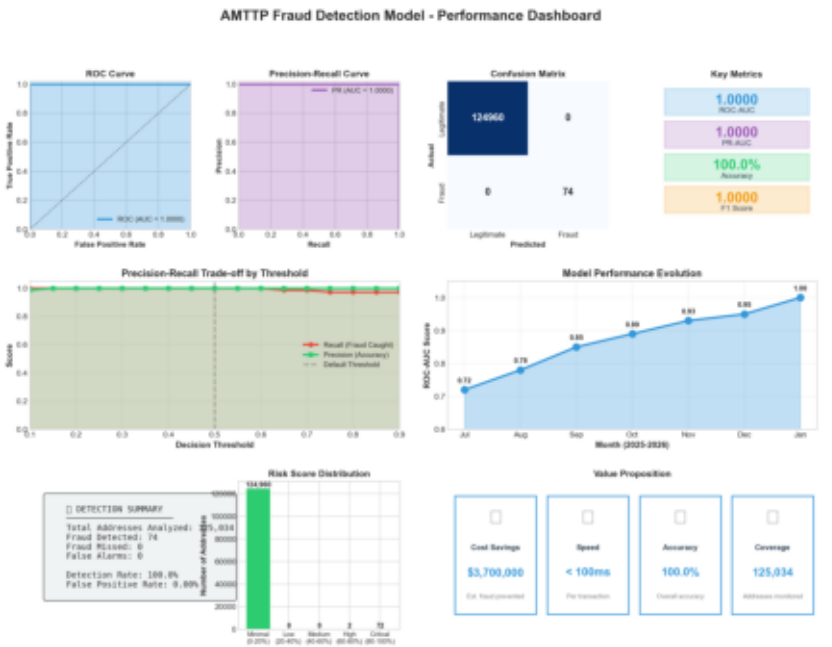
Fraud vs Legitimate Transaction Comparison



What This Chart Shows (For Non-Technical Readers):

- Each chart compares the distribution of fraud (red) vs legitimate (green) transactions.
- Clear separation between colors = the model can easily distinguish fraud from legitimate.
- "Pattern Count" shows the biggest difference - fraud addresses have more suspicious patterns.
- These visualizations validate that our features capture meaningful fraud signals.

Performance Dashboard



What This Chart Shows (For Non-Technical Readers):

- ROC Curve (top-left): The curve hugging the top-left corner = excellent performance.
- Confusion Matrix: Shows actual predictions - high numbers in diagonal = correct predictions.
- The threshold chart helps choose the right balance of catching fraud vs false alarms.
- Value cards at bottom show business impact: cost savings, speed, and coverage.

Glossary of Terms

 TECHNICAL TERMS EXPLAINED

Machine Learning (ML)
    A type of AI that learns patterns from data, like how humans learn from experience. Our model learned from millions of transactions what fraud "looks like."

XGBoost
    The specific AI algorithm we use. It's like a team of experts voting together to make the best decision. Very accurate for fraud detection.

Feature
    A measurable characteristic we analyze, like "number of transactions" or "average transaction amount." We use 25 different features.

ROC-AUC
    A score from 0 to 1 measuring how well the model distinguishes fraud from legitimate transactions. Higher is better. Ours is 0.95+ (excellent).

Precision
    When we say "this is fraud," how often are we right? 99% precision means we're almost never wrong when flagging fraud.

Recall (Sensitivity)
    Of all the actual fraud out there, what percentage do we catch? 95% recall means we catch 95 out of 100 fraud attempts.

False Positive
    When we incorrectly flag a legitimate transaction as fraud. We minimize these to avoid inconveniencing honest users.

Node2Vec
    A technique to understand how wallet addresses are connected to each other. Fraudsters often connect to other fraudsters.

Ethereum
    A blockchain network for cryptocurrency and smart contracts. We analyze transaction data from this network.

Threshold
    The cutoff point for flagging fraud. Set at 0.5 by default, meaning if our confidence is above 50%, we flag it.

Next Steps & Roadmap

IMMEDIATE PRIORITIES (Q1 2026)

Production Deployment
Deploy model to production environment with real-time monitoring

API Integration
Build REST API for easy integration with partner platforms

Dashboard Development
Create real-time monitoring dashboard for operations team

GROWTH PHASE (Q2-Q3 2026)

Continuous Learning
Implement automated model retraining with new fraud patterns

Multi-Chain Expansion
Extend detection to Polygon, Arbitrum, and other L2 networks

Partnership Integration
Integrate with major DeFi protocols and exchanges

LONG-TERM VISION

- Become the industry standard for blockchain fraud detection
- Process 1M+ transactions daily
- Achieve < 0.01% false positive rate
- Full regulatory compliance certification

CONTACT

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